Chapter 4, Section 6

James Lee

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1. A rational curve of degree 4 in $\mathbb{P}^3$ is contained in a unique quadric surface Q, and Q is necessarily nonsingular. A

Proof. Let X be a rational curve of degree 4 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow H^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Now the middle vector space has dimension 10, and $\mathcal{O}_X(2)$ corresponds to a linear system of degree 8, so it has dimension at most 9 by Riemann-Roch. Then, $H^0(\mathbb{P}^3, \mathcal{I}(2))$ has dimension at least 1, so X is contained in a quadric surface Q. If X is contained in another quadric surface, then it must be a complete intersection by reason of its degree, which implies $\omega_X = \mathcal{O}_X$ (II, Ex. 8.4), a contradiction. If Q is singular, then X must have genus 1 (6.4.1), which is a contradiction. Hence, Q must be a nonsingular quadric surface. □

2. A rational curve of degree 5 in $\mathbb{P}^3$ is always contained in a cubic surface, but there are such curves which are not contained in any quadric surface.

Proof. Let X be a rational curve of degree 5 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(3)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{I}(3)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(3)) \longrightarrow H^0(X, \mathcal{O}_X(3)) \longrightarrow \cdots.$$

Now, the middle vector space has dimension 20, and the right hand vector space has dimension 16 by Riemann-Roch on X (note that $\mathcal{O}_X(3)$ corresponds to a divisor of degree 15). So we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(3)) \geqslant 4$.

□

3. A curve of degree 5 and genus 2 in $\mathbb{P}^3$ is contained in a unique quadric surface Q. Show that for any abstract curve X of genus 2, there exist embeddings of degree 5 in $\mathbb{P}^3$ for which Q is nonsingular, and there exist other embeddings of degree 5 for which Q is singular.

Proof. Let X be a curve of degree 5 and genus 2 in $\mathbb{P}^3$, and let $\mathcal{I}$ be its ideal sheaf. Then we have an exact sequence

$$0 \longrightarrow \mathcal{I} \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0.$$

Twisting by $\mathcal{O}_{\mathbb{P}^3}(2)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow H^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

The middle vector space has dimension 10, and the right hand vector space has dimension 9 by Riemann-Roch on X (note that $\mathcal{O}_X(2)$ corresponds to a nonspecial divisor of degree 10). So we conclude that $h^0(\mathbb{P}^3, \mathcal{I}(2)) \geqslant 1$. By reason of degree, X cannot be contained in two distinct quadric surfaces, so there is a unique irreducible quadric surface Q containing X.

□

4. There is no curve of degree 9 and genus 11 in $\mathbb{P}^3$.

Proof. Suppose there exists a curve $X \subseteq \mathbb{P}^3$ of degree 9 and genus 11, and let $\mathcal{I}$ be its ideal sheaf. We have an exact sequence

$$0 \longrightarrow H^0(\mathbb{P}^3, \mathcal{I}(2)) \longrightarrow H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(2)) \longrightarrow H^0(X, \mathcal{O}_X(2)) \longrightarrow \cdots.$$

Then $\mathcal{O}_X(2)$ is a very ample line bundle corresponding to a linear system of degree 18, say $|\mathcal{O}_X(2)| = |D|$ for some very ample divisor D of degree 18. Riemann-Roch gives

$$\dim |D| - \dim |K - D| = 8.$$

If D is nonspecial, then $\dim |D| = 8$. If D is special, Clifford's theorem (5.4) says

$$\dim |D| \leqslant \frac{1}{2} \deg D = 9,$$

and we have an equality if and only if X is hyperelliptic and D is a multiple of the unique g^1_2 on X. But D is very ample, so it cannot be a multiple of the g^1_2, and the inequality must be strict. We conclude that

$$h^0(X, \mathcal{O}_X(2)) = 1 + \dim |D| \leqslant 9,$$

and therefore, $H^0(\mathbb{P}^3, \mathcal{I}(2)) \neq 0$. Hence, X must be contained in a quadric surface.

Following the discussion in (6.4.1), if X is contained in nonsingular quadric surface, then it corresponds to a divisor of type (a, b) such that

$$a + b = 9, \quad ab - a - b + 1 = 11,$$

which is not possible. If X is contained in a singular quadric surface, then X must have genus 12. Hence, there is no curve of degree 9 and genus 11 in $\mathbb{P}^3$. □

5. If X is a complete intersection of surfaces of degree a, b in $\mathbb{P}^3$, then X does not lie on any surface of degree $< \min(a, b)$.

Proof. If X is a complete intersection of surfaces H, H' of degree a, b respectively, in $\mathbb{P}^3$, then X has degree ab. Without loss of generality, suppose $a \leqslant b$. If H'' is surface of degree $c < a$ containing X, then the irreducible components of $H \cap H''$ can have degree at most ac (I, 7.7), which is strictly less than ab, so X cannot be an irreducible component of $H \cap H''$. Hence, X cannot lie on any surface of degree $< a$. □